

AI-Powered Question Bank Generator

Final Year Project Synopsis

Abstract

This project automatically generates 2-mark, 5-mark, and 10-mark questions from uploaded syllabus notes or PDF documents. Using Natural Language Processing (NLP) and Large Language Models (LLMs), the system extracts key concepts, identifies important topics, and creates question banks, model answers, and quizzes for students and faculty.

Objectives

- Automate question paper preparation.
- Generate questions of different difficulty levels.
- Produce model answers and quizzes.
- Reduce faculty workload and improve student preparation.

Modules

1. PDF Upload Module
2. Text Extraction Module
3. Content Analysis Module
4. Question Generation Module
5. Answer Generation Module
6. Quiz Generator
7. Admin Dashboard

Technology Stack

Frontend: HTML, CSS, JavaScript/React

Backend: Python Flask/FastAPI

Database: MySQL/SQLite

AI/NLP: Transformers, LangChain, Gemini/OpenAI APIs

PDF Processing: PyMuPDF, pdfplumber

Expected Outcome

The system generates structured question banks, model answers, and quizzes from uploaded academic materials, helping both students and educators.